

Introduction to Machine Learning

Machine learning is a branch of artificial intelligence that enables systems to learn from data and improve their performance without being explicitly programmed.

There are three main types of machine learning: supervised learning, unsupervised learning, and reinforcement learning.

In supervised learning, the model is trained on labelled data - each input has a corresponding correct output. Common examples include email spam detection, image classification, and house price prediction.

Neural Networks

A neural network is a computational model loosely inspired by the human brain. It consists of layers of interconnected nodes called neurons. The first layer is the input layer, the last is the output layer, and everything in between are hidden layers.

Deep learning refers to neural networks with many hidden layers. These deep networks excel at tasks like image recognition, speech recognition, and natural language processing. GPT and BERT are examples of deep learning models.

Large Language Models

Large Language Models (LLMs) are neural networks trained on vast amounts of text. They learn to predict the next word in a sequence, which forces them to build rich internal representations of language and knowledge.

GPT-4, Claude, and Gemini are examples of LLMs. They are trained on trillions of tokens and contain billions of parameters. Fine-tuning allows these general models to be specialised for specific tasks such as coding or medical advice.

Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines a retrieval system with a language model. Instead of relying solely on the LLM's memorised knowledge, RAG fetches relevant documents at query time and provides them as context.

The pipeline has three stages: first, documents are split into chunks and embedded into a vector space. Second, when a question arrives it is embedded and the nearest chunks are retrieved. Third, the chunks and question are sent to the LLM which generates a grounded answer.

RAG reduces hallucination, allows the model to cite sources, and lets you update the knowledge base without retraining the model.

Vector Databases

A vector database stores high-dimensional vectors and supports fast similarity search. Popular choices include ChromaDB, Pinecone, Weaviate, and Qdrant.

The most common indexing algorithm is HNSW (Hierarchical Navigable Small World). HNSW builds a graph where nearby vectors are connected, enabling sub-linear search time. This is critical when the knowledge base contains millions of chunks.

Cosine similarity and dot product are the most common distance metrics used to rank retrieved results by relevance.